

AE646 — Individual Contribution Statement & AI Tool Use Declaration

Team PINNacles

Aryamann Srivastava

November 2026

Individual Contribution Statement

Team member: Aryamann Srivastava

- **Problem formulation and literature review:** Identified the 2D Darcy flow benchmark in PDEBench as the project focus; reviewed the FNO paper [1] and PDEBench paper [2].
- **Data pipeline:** Implemented `src/download_data.py` and `src/preprocess.py` for downloading and preprocessing PDEBench data; designed the 900/100/200 train/val/test split (a 1000-sample train/val pool plus 200 held-out test samples, drawn with a fixed seed from PDEBench's 10000-sample archive).
- **Model implementation:** Implemented `MLPBaseline`, `SpectralConv2d`, and `FNO2d` in `src/models.py`; verified correctness against the reference FNO codebase.
- **Training:** Implemented `src/train.py` with AdamW and cosine LR annealing, retaining the checkpoint with the lowest validation relative- L_2 error; ran all three training runs (FNO original, MLP, FNO improved) on the lab GPU VM.
- **Evaluation:** Implemented `src/evaluate.py`, `src/superres_eval.py`, and `src/benchmark_speed.py`; collected all quantitative results.
- **Component profiling:** Designed and implemented `src/benchmark_components.py` with correct GPU synchronisation to measure per-layer execution time.
- **Figure generation:** Implemented `src/generate_figures.py`; produced all six report figures from stored JSON results.
- **Reports and presentations:** Wrote Stage 1 proposal (`stage1/PINNacles_Stage1_Proposal.tex`), Stage 1 presentation (10 slides), Stage 2 interim report (`stage2/PINNacles_Stage2_InterimReport.tex`), the Stage 3 final presentation (`PINNacles_Stage3_FinalPresentation`), and this final report (`stage3/PINNacles_Stage3_FinalReport.tex`).

Team PINNacles was formed with four members at the proposal stage (Aryamann Srivastava, Varun Sathaye, Atishay Jain, Vedant Shekhar Tiwari). The implementation, experiments, figures, and reports contained in this repository were carried out by Aryamann Srivastava, whose individual contributions are listed above.

AI Tool Use Declaration

In accordance with the AE646 course policy on generative AI tools, I declare the following use of AI assistance during this project:

1. **Code debugging:** Claude Code (Anthropic) was used to identify and fix a GPU timing bug in the component profiling script. The issue was that `time.perf_counter()` measured dispatch time rather than execution time for asynchronous GPU operations; the fix (a device-synchronisation call before every timer read — `torch.cuda.synchronize` on CUDA, `torch.mps.synchronize` on Apple Silicon) was suggested by the AI assistant. All code was reviewed, understood, and accepted by me before being committed.
2. **Report drafting and consistency checking:** Claude Code assisted with LaTeX structure and table formatting, with drafting and editing report prose, and with cross-checking the numbers quoted in the reports against the stored results files (`results/*/eval_metrics.json`, `benchmark_*.json`) and the training configs. Every number originates from my own experiments; all scientific claims and their interpretation are my own and were verified by me.
3. **Presentation slides:** Claude Code assisted with Python code to programmatically build the Stage 1 PPTX presentation using `python-pptx`. The technical content of every slide was written by me.
4. **No AI-generated results:** All numerical results in this report are from experiments I ran on real hardware. No results were generated or hallucinated by AI tools.

Signature: Aryamann Srivastava

Date: November 2026

References

- [1] Z. Li et al., “Fourier Neural Operator for Parametric Partial Differential Equations,” *ICLR*, 2021.
- [2] M. Takamoto et al., “PDEBench,” *NeurIPS*, 2022.